Assignment 4

Introduction

In previous assignments, you focused on **Exploratory Data Analysis**, **Data Cleaning**, and **Classical Machine Learning techniques**.
In this assignment, the focus shifts toward **modern data science workflows**, including:

- Deep Generative Models
- Handling Imbalanced Data
- Explainable AI (XAI)
- End-to-end ML Pipelines

These topics form the foundation of many real-world machine learning systems used in industry and research.

In this homework, you will study and implement the following concepts:

- Variational Autoencoders (VAEs)
- Generative Adversarial Networks (GANs)
- Diffusion Models
- Machine Learning Pipelines
- Handling Imbalanced Data
- Explainable AI (XAI)

Submission Guidelines

Your results must be presented in a well-documented **Jupyter Notebook (** `.ipynb` **)**.

- Keep your notebook **clean, modular, and reproducible**
- The focus is on **clarity and understanding**, not quantity or excessive training

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How to Submit

1. A standalone Jupyter notebook, or
2. Use the **same GitHub repository** you used for previous homework
3. You may also share your notebook via **Google Colab** (ensure link access is open)

4. You may organize your code modularly (e.g., `.py` files imported into the notebook)

Collaboration Policy

All homeworks must be done **individually**.

Evaluation Criteria

You will be graded **qualitatively**, based on:

- Correct and appropriate implementation of models and pipelines
- Clear, readable, and well-commented notebooks
- Meaningful explanations and interpretations of experiments
- Analysis of how results change when:
  - Data processing steps are modified
  - Modeling choices are altered
  - Experimental settings are adjusted
- Avoiding unnecessary training or excessive computation
- Effective use of figures, tables, and comments to support understanding

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Late Submission Policy

A **10% penalty** will be applied for **each late day**.

Generative AI Policy

The use of tools such as **ChatGPT, Claude, Gemini**, or similar AI assistants is **allowed and encouraged**, but use them wisely.

- Try to solve each problem yourself first
- Then, use AI tools to check, improve, or compare your solutions
- What matters most is **understanding the code you submit**, not who wrote it
- Be cautious — large language models may *hallucinate* or produce inaccurate results

Important:
It is strongly recommended that you use the course's **AI Teaching Assistant** before the deadline and upload:

- Your answers
- Approximate scores
- Suggestions for improving your implementations

[AI ADS Assistant]

The goal is to help you become confident in solving real-world data science problems **independently**.

Homework Components

Part 1: ML Pipelines

Using the dataset from **Assignment 2 or 3** where class imbalance exists:

A. Data Cleaning Pipeline (Pandas)

- Create a data loading and cleaning pipeline using `pandas.pipe`
- Implement custom functions for data validation
- Chain multiple cleaning operations together

B. Preprocessing Pipeline (Scikit-learn)

- Build preprocessing pipelines using `sklearn.pipeline`
- Include appropriate feature scaling
- Handle categorical feature encoding within the pipeline

C. Missing Data Handling

- Use an **Imputer** inside your pipeline
- Compare different imputation strategies:
  - Mean
  - Median
  - KNN
  - Iterative imputation

D. End-to-End Pipeline

- Attach a classifier to the pipeline

- Ensure the entire pipeline is trainable as a single unit
- Save and load the complete pipeline

Part 2: Handling Imbalanced Data

Using the dataset from **Assignment 1** with imbalanced classes:

A. Random Undersampling

- Apply random undersampling to majority classes
- Compare different undersampling strategies
- Analyze the impact on model performance

B. Random Oversampling

- Apply random oversampling to minority classes
- Compare results with undersampling

C. SMOTE

- Implement **SMOTE (Synthetic Minority Over-sampling Technique)**
- Explain how SMOTE generates synthetic samples
- Compare SMOTE with simple oversampling

D. Cost-Sensitive Learning

- Compute appropriate class weights
- Integrate class weights into your classifier
- Analyze performance differences

Part 3: Variational Autoencoders (VAEs)

A. VAE Implementation

- Implement and train a **Variational Autoencoder**
- Use an image dataset of your choice

B. KL-Divergence Exploration

- Experiment with different KL-divergence weights (**β-VAE**)
- Analyze the effect on:
  - Latent disentanglement
  - Image quality

C. Latent Space Sampling

- Interpolate between two latent vectors
- Generate and visualize new samples

Part 4: Generative Adversarial Networks (GANs)

Using the **MNIST dataset**, implement a GAN:

- Implement the **Generator**
- Implement the **Discriminator**
- Implement the training loop with appropriate loss functions
- Generate and visualize samples across training epochs

Discussion Question

- Why can GANs potentially suffer from **mode collapse**?
- Why is there no formal guarantee that mode collapse will occur?

Part 5: Diffusion Models

A. Denoising Diffusion Model

- Implement a simple **DDPM** for image generation

B. Noise Schedule Experimentation

- Compare **linear** and **cosine** noise schedules
- Analyze their effect on sample quality

C. Diffusion vs. GAN Comparison

- Compare diffusion models and GANs in terms of:
  - Output quality
  - Diversity
  - Training stability

Part 6: Explainable AI (XAI)

Using the convolutional architecture from **Assignment 3**, analyze model behavior:

A. Visual Explanations

- Implement **Grad-CAM**
- Generate heatmaps for:
  - Correctly classified images
  - Misclassified images
- Explain what the model is focusing on

B. Model-Agnostic Explainability

Implement and analyze:

- **SHAP**
  - Summary plots
  - Waterfall plots
- **LIME**
  - Superpixel-based explanations
- **ELI5**
  - Feature importance visualizations

Analysis and Insights

For **at least 3 misclassified images**, provide a detailed analysis:

- What features did the model focus on?
- Why did the misclassification occur?
- What could be improved in the training process?

Optional Bonus (End-to-End Workflow)

Choose a single dataset and apply the **full pipeline**:

Data Cleaning → ML Pipeline → Imbalance Handling → Generative Model → XAI

- Compare results **before and after** each stage
- Discuss improvements, trade-offs, and limitations

Contact & Questions

If you have any questions, feel free to ask in the **Bale group**.

IMPORTANT NOTE: In your notebook, per cell, please explain why you are doing that part (in natural language, Farsi or English). Also, you need to explain what you have gained/understood from that part. If you only provide code without the comments, **you will not get the full mark**.

Good Luck :)

Due date: Fri, Tir 12, 23:59:59